

Solving Linear Equations

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Elimination is a systematic way to solve linear equations. It removes one unknown at a time until the system is upper triangular; back substitution then works from the bottom upward. Each elimination step changes the equations without changing their common solution.

Adapted from Gilbert Strang, Section 2.2, pp. 46–47.

SYSTEMS AND ELIMINATION

$$Ax = b, \quad Ax = x_1 a_1 + \cdots + x_n a_n.$$

Row operations:

interchange rows; multiply a row by a nonzero constant; add a multiple of one row to another.

Rank criterion:

$Ax = b$ is solvable iff $\text{rank}(A) = \text{rank}([A \ b])$.

Pivot theorem:

free variables occur in non-pivot columns; if r is the rank, the solution set has $n - r$ free parameters.

$$Ax = 0 \Rightarrow x = x_{\text{pivot}} + x_{\text{free}}; \quad x = x_p + x_n, \quad Ax_n = 0.$$

Rouché–Capelli conclusion:

a consistent system has either one solution (no free variables) or infinitely many solutions.

MATRIX OPERATIONS

$$A^{-1}A = I, \quad (AB)^{-1} = B^{-1}A^{-1}, \quad (A^T)^{-1} = (A^{-1})^T. \\ E_{ij}^{-1} = E_{ij}(-c), \quad E_i(c)^{-1} = E_i(c^{-1}), \quad P^{-1} = P^T.$$

Elimination factorization:

$EA = U$, hence $A = LU$; with row exchanges, $PA = LU$.

LU solution:

solve $Ly = b$ followed by $Ux = y$.

SPECIAL MATRICES

$$A^T A \text{ is symmetric, } A = S + K, \quad S = \frac{A + A^T}{2}, \quad K = \frac{A - A^T}{2}.$$

Orthogonal matrix:

$Q^T Q = I \iff Q^{-1} = Q^T$; it preserves dot products and lengths.

$$\det(AB) = \det A \det B, \quad A^{-1} = \frac{1}{\det A} \text{Cof}(A)^T.$$

PROBLEM SET

Q1.

Consider the system

$$2x + 3y = 1, \quad 10x + 9y = 11.$$

- (i) Use elimination to obtain an upper triangular system and identify its two pivots.
- (ii) Solve the system by back substitution and verify the solution as a linear combination of the columns of the coefficient matrix.
- (iii) Without repeating the full elimination, solve the system when the right side is changed to $(4, 44)^T$.

Q2.

Suppose that $Ax = b$ has two distinct solutions x_1 and x_2 .

- (i) Show that $(1 - t)x_1 + tx_2$ is also a solution for every $t \in \mathbb{R}$.
- (ii) Deduce that a system of linear equations cannot have exactly two solutions.

Q3.

Let

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 4 & 6 & 1 \\ -2 & 2 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

- (i) Find the elimination matrices E_{21} , E_{31} , and E_{32} that reduce A to an upper triangular matrix U .
- (ii) Compute $M = E_{32}E_{31}E_{21}$ and verify that $MA = U$.
- (iii) Apply the same elimination steps to b and solve $Ax = b$ by back substitution.

Q4.

Let

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}.$$

- (i) Compute AB and BA , and show that matrix multiplication is not commutative in this example.
- (ii) Expand $(A + B)^2$ without assuming that A and B commute.
- (iii) Compute $(A + B)^2$ and $A^2 + 2AB + B^2$. Explain precisely when the scalar-style identity $(A + B)^2 = A^2 + 2AB + B^2$ is valid for square matrices.

Q5.

Let A be a 3×3 matrix.

- (i) Suppose that the third row of A is the sum of its first two rows. Show that A is not invertible. Derive a necessary condition on $b = (b_1, b_2, b_3)^T$ for $Ax = b$ to be consistent, and describe what happens to the third equation during elimination.
- (ii) Suppose instead that the third column of A is the sum of its first two columns. Find a nonzero solution of $Ax = 0$, and use it to show that A is not invertible.

Q6.

Use Gauss-Jordan elimination on $[A \ I]$ to find A^{-1} for

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Check your answer by matrix multiplication.

Q7.

Let

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 7 \\ 11 \end{bmatrix}.$$

- (i) Use elimination without row exchanges to factor $A = LU$, where L has ones on its diagonal.
- (ii) Solve $Lc = b$ by forward substitution and then solve $Ux = c$ by back substitution.

Q8.

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$.

- (i) Prove from the entries of the matrices that $(AB)^T = B^T A^T$.
- (ii) Let $Q = [q_1 \ \cdots \ q_n]$ be an $n \times n$ matrix satisfying $Q^T Q = I$. Show that the columns of Q are unit vectors and are mutually orthogonal. Prove the converse.
- (iii) Show that Q preserves dot products and lengths. Give a 2×2 example with first entry $q_{11} = \cos \theta$.